

RAISA: Multi-Agent Research Assistant

Dr. Venkata Rami Reddy Chirra
Guide, SCOPE
VIT-AP University, Amaravati, India
venkataramireddy.chirra@vitap.ac.in

Sk.Zaafira Yumn
Student, SCOPE
VIT-AP University, Amaravati, India
zaafira.23bce9719@vitapstudent.ac.in

M.Narendra Kumar
Student, SCOPE
VIT-AP University, Amaravati, India
narendra.23bce20218@vitapstudent.ac.in

Abstract—The proliferation of scholarly articles has rendered the process of literature review more difficult and tedious than ever before. Existing research frameworks mainly focus on article retrieval and indexing, lacking advanced capabilities such as summarization, trend analysis, and research gap identification.

In this paper, we introduce *RAISA*, a multi-agent system powered by Artificial Intelligence and Natural Language Processing that automates the research workflow. RAISA combines article search with the use of the OpenAlex API, keyword extraction with the help of KeyBERT, semantic analysis with Sentence Transformers, and article summarization with artificial intelligence.

Moreover, it conducts trend analysis and searches for any research gaps, producing well-structured PDF reports to facilitate decision-making. Our experimental findings indicate that RAISA is quite effective at decreasing the effort required for research. Experimental evaluation demonstrates that RAISA significantly improves efficiency and reduces manual effort compared to traditional research methods.

Index Terms—Research Assistant, Multi-Agent Systems, Natural Language Processing, Artificial Intelligence, Research Gap Detection, Keyword Extraction, Text Summarization, OpenAlex API

I. INTRODUCTION

The rise in the number of scholarly articles published in the recent years has made it very difficult to perform efficient literature reviews. Scholars have to read through numerous documents and gather information from them. They also need to identify the research gaps, which requires considerable effort. Existing research frameworks like Google Scholar and Semantic Scholar have mainly concentrated on the searching and indexing of papers rather than analysis. However, while there have been remarkable developments in areas such as AI and NLP, which include the ability to extract keywords, comprehend semantics, and summarize texts, such features are offered as separate utilities. The failure to integrate such utilities leads to fragmented processes, where researchers have to depend on several applications to obtain significant insights.

In this regard, this study introduces **RAISA (Research Assistant Intelligent Software Agent)** as a multi-agent research assistant to automate the entire research workflow. The research assistant integrates various utilities, including data acquisition, natural language processing, and analytics frameworks. The software utilizes the OpenAlex API to collect

papers, KeyBERT to identify keywords, Sentence Transformers for semantics, and AI for summaries. Apart from these abilities, it has a feature for conducting trend analysis for identifying publication trends, along with incorporating a lightweight research gap detection algorithm based on keyword distribution. The RAISA framework is also able to produce structured PDF reports to allow users to easily get information from vast academic literature.

The main contributions of this work are as follows:

- A novel multi-agent architecture for automated research analysis
- Integration of Artificial Intelligence and Natural Language Processing for end-to-end processing
- Automated research gap detection and trend analysis
- Generation of structured and interpretable research reports

The structure of the rest of this document is as follows: Section II outlines previous work in the field, Section III is the literature review section, Section IV proposes the system design, Section V provides the methodology, Section VI deals with implementation, Section VII provides the results, and Section VIII summarizes the whole paper.

II. RELATED WORK

There have been some developments of frameworks that help researchers to access and analyze academic papers. Frameworks like Google Scholar and Semantic Scholar provide functionality related to finding articles and their citation analysis. But these frameworks focus mainly on searching and indexing capabilities and cannot be used for further analysis.

Nowadays, advances in natural language processing technology lead to development of various applications like KeyBERT, used to extract keywords from papers, and SentenceBERT that can recognize semantic meaning of sentences. Nevertheless, these techniques do not belong to any process, remaining isolated solutions.

Manual approach to research literature analysis remains one of the main disadvantages of traditional literature reviews as it requires a lot of time and labor.

However, until now there was no platform that would unite all processes of literature analysis – from research finding to generation of insights. That is why it makes sense to develop the proposed system.

Recent studies have explored the use of Natural Language Processing techniques for automating literature review tasks. However, these approaches are often limited to specific functionalities and lack a unified framework that integrates multiple research assistance capabilities into a single system.

III. LITERATURE REVIEW

Table I shows the differences between existing research systems according to their methodology, attributes, and shortcomings. While conventional systems, including Google Scholar, concentrate on document search and citation indexing, Semantic Scholar offers artificial intelligence-driven recommendations but is not efficient in research gaps discovery. Natural language processing-based methods include document summarization, key phrase extraction, and sentence embedding; however, they are designed as separate systems with no research cycle integration. From the comparison, it is evident that existing systems either focus on search and retrieval or provide isolated analytical capabilities. None of the systems offer a fully integrated solution that combines data retrieval, analysis, and report generation. RAISA addresses this limitation by providing a unified multi-agent framework.

IV. PROPOSED SYSTEM

The proposed system, RAISA (Research Assistant Intelligent Software Agent), is designed as a multi-agent framework to automate the research workflow. It integrates Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques to assist users in retrieving, analyzing, and summarizing academic literature efficiently.

A. System Overview

RAISA operates by accepting a user-defined research query and processing it through a sequence of intelligent agents. The system retrieves relevant research papers, performs NLP-based analysis, identifies trends, detects research gaps, and generates structured reports. This automation significantly reduces manual effort and improves research productivity.

B. System Architecture

The architecture of RAISA is shown in Fig. 1. The system is composed of multiple agents, each responsible for a specific function within the research pipeline.

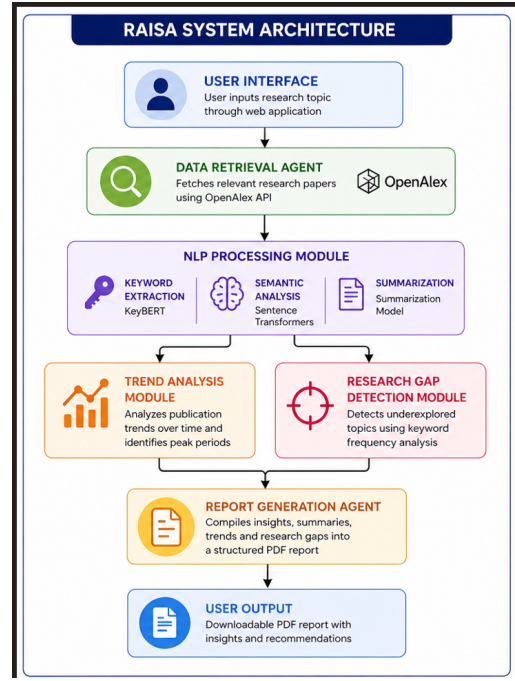


Fig. 1: Proposed system Architecture

The major components of the system include:

- **User Interface:** Accepts research queries from the user.
- **Data Retrieval Agent:** Fetches research papers using the OpenAlex API.
- **Processing Agent:** Performs keyword extraction, summarization, and semantic analysis using NLP techniques.
- **Trend Analysis Module:** Identifies publication patterns over time.
- **Research Gap Detection Module:** Detects underexplored areas using keyword frequency analysis.
- **Report Generation Agent:** Generates structured PDF reports.

C. Workflow

The workflow of RAISA is illustrated in Fig. 2.

The workflow consists of the following steps:

- 1) The user inputs a research topic.
- 2) The system retrieves relevant papers using the OpenAlex API.
- 3) NLP models extract keywords and generate summaries.
- 4) Trend analysis and research gap detection are performed.
- 5) A structured report is generated and presented to the user.

D. Module Description

RAISA consists of multiple modules working collaboratively:

- **Data Collection Module:** Retrieves research data from external APIs.
- **NLP Processing Module:** Performs keyword extraction and summarization.

TABLE I: COMPARATIVE ANALYSIS OF EXISTING RESEARCH SYSTEMS

System	Technique	Key Features	Limitations	RAISA Advantage
Google Scholar	Search	Ranking	Paper retrieval, citation indexing	No AI-based analysis
Semantic Scholar	AI Search	Recommendations, citation graphs	Limited gap detection	Identifies trends and gaps
KeyBERT	Transformer	Keyword extraction	Not a complete pipeline	Integrated into full workflow
Sentence-BERT	Embeddings	Semantic similarity, text understanding	No direct research analysis	Enables deeper insights
Manual Methods	Traditional	Detailed literature review	Time-consuming, not scalable	Fully automated process
RAISA	Multi-Agent	End-to-end automation, report generation	Depends on external APIs	Complete intelligent system



Fig. 2: RAISA Workflow

- **Analysis Module:** Conducts trend analysis and identifies research gaps.
- **Storage Module:** Stores user data and research results using SQLite.
- **Output Module:** Generates downloadable PDF reports.

V. METHODOLOGY

This sub-section explains the methods and processes involved in automating research analysis through the proposed system. It is designed to integrate NLP, machine learning algorithms, and data search methods to enable automatic research assistance.

A. Data Collection

The proposed system searches for research articles by connecting to the OpenAlex API, which has access to numerous scholarly articles. According to the input query from the user, research articles are searched, along with their metadata such as title, abstract, publication date, and citations.

B. Keyword Extraction

Keyword extraction is done by **KeyBERT**. It uses contextual embeddings to detect and extract relevant keywords from the text. This enables efficient identification of key concepts within research papers.

C. Semantic Analysis

A deeper semantic analysis is performed by **Sentence Transformers (Sentence-BERT)**. Sentence transformers generate embeddings for sentences. It facilitates a thorough comparison of research papers by finding similarities among them.

D. Text Summarization

The process of text summarization is also carried out using AI technologies by RAISA. This enables researchers to quickly understand the key insights from research papers.

E. Trend Analysis

This analysis process involves examining the number of publications over time. The framework identifies the prevalence of the research topic over varying years to determine emerging trends and declining trends in the domain.

F. Research Gap Identification

The research gap is determined by applying the frequency of keywords and semantic analysis on the selected research topic. Research gaps are usually determined when the topic receives little attention or low frequency from keyword usage.

G. Output Generation

The output will be provided in a PDF file format. The generated report contains summarization results, keyword analysis, trend analysis, and research gaps within the literature.

H. System Integration

RAISA employs the Flask framework for its backend component to perform processing operations and communicate with other APIs. The front-end utilizes HTML, CSS, and JavaScript. Data is stored in SQLite, while ReportLab is responsible for producing a report in the PDF format.

I. Approach of Algorithm

The proposed system adopts an organized pipeline for handling research queries. At the outset, the user queries will be utilized to obtain relevant research articles from the OpenAlex API. Then, the abstracts of the documents obtained will be used to extract keywords using KeyBERT and generate semantic embeddings using Sentence Transformers.

The generated keywords are evaluated in terms of frequency for identifying trends and detecting gaps in research. Semantic similarity score will also be used to determine connections between research areas.

The combination of these techniques ensures an efficient and automated research analysis pipeline.

VI. IMPLEMENTATION

This section discusses the implementation of the RAISA framework, including its technological aspects and the integration of various components in order to automate research.

A. System Design

RAISA uses a client-server framework for implementation. The server-side of the application is written using Flask and is responsible for handling API requests, performing data manipulation, and enabling inter-module communication. The client-side interface of the application is created using HTML, CSS, and JavaScript.

B. Data Retrieval Implementation

The OpenAlex API is used by the RAISA framework to dynamically retrieve research papers. When the user enters a search query, an API request is made to retrieve papers along with their metadata, including title, abstract, publication year, and citations.

C. Implementation of NLP Module

The NLP module is implemented with transformer-based models as follows:

- **KeyBERT:** Keyword extraction from research abstracts.
- **Sentence Transformers:** Semantic embeddings generation and analysis.
- **Summarization Models:** Paper summary generation.

This enables the system to effectively process and analyze textual data.

D. Trend Analysis and Gap Identification

Trend analysis can be implemented by grouping papers according to their publication years and analyzing the distribution of keyword frequencies. The identification of research gaps can be done through identifying underrepresented keywords.

E. Database and Storage

SQLite is chosen as the database solution for storing user inputs, retrieved articles, and generated outputs. This will facilitate data management and fast access to pre-analyzed data.

F. Reports

The software creates reports in PDF format, using the ReportLab module. Reports include information about the main points, keywords, trends, and knowledge gaps. This capability lets users save and use output files for their research purposes.

G. Integration and Deployment

All modules are combined into one software package. The backend processes input data and applies models, whereas the frontend serves as an interface for end-users. This program can be run either locally or in the cloud environment.

To summarize, the development of RAISA provides smooth implementation of research activities with the help of AI and web technologies.

H. User Interface

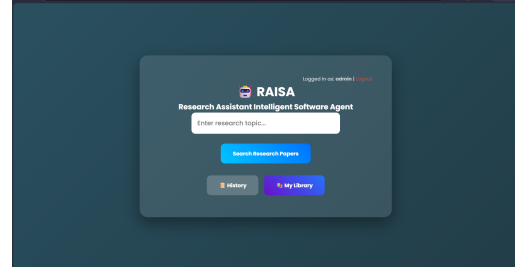


Fig. 3: RAISA user interface for entering research queries

VII. RESULTS AND DISCUSSION

In this section, we present the output of the proposed system and analyze the efficiency of the proposed solution in automating research analysis.

A. System Output

The proposed system efficiently searches for relevant research papers in the user's request by leveraging the OpenAlex API. After retrieving papers, the system extracts their title, abstract, year, and citation number.

The outputs of the system are keyword extraction, content summarization, and structured insights generation. These outputs enable users to quickly understand key insights from multiple research papers without manual analysis. In addition, the system creates a structured report in PDF format.

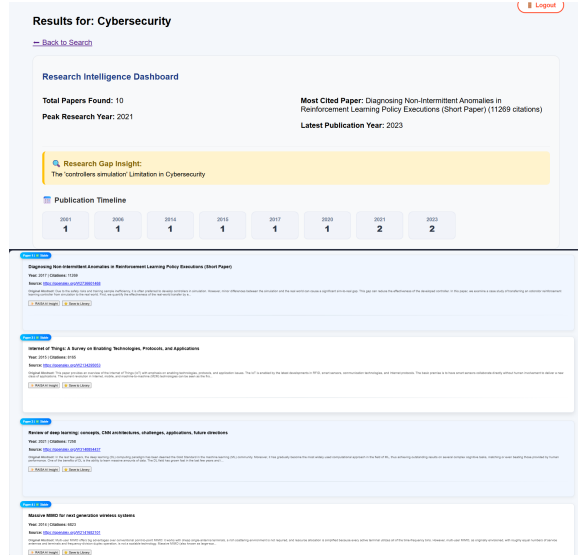


Fig. 4: RAISA dashboard displaying research insights including paper statistics, trends

B. Dashboard Analysis

In this section, we analyze the dashboard's output and evaluate its usefulness for understanding the retrieved research papers. This output consists of several important pieces of information, such as the total number of papers, peak year, and the most cited paper.

In addition, the visualization of publications allows one to understand how the research area evolves over time. The system also identifies research gaps for further exploration.

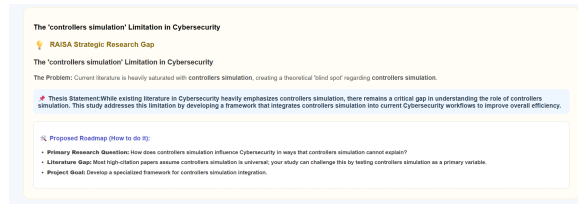


Fig. 5: RAISA dashboard showing Research Gap

C. Evaluation Metrics

The performance of the proposed system is evaluated using efficiency, output quality, and system performance metrics.

1) Efficiency: The system significantly reduces the time required for literature review by automating data retrieval, analysis, and report generation.

2) Output Quality: The relevance of retrieved papers and the clarity of generated summaries are high, ensuring meaningful and accurate insights.

3) System Performance: RAISA demonstrates stable performance with consistent API responses and low processing latency.

4) Functional Effectiveness: The system effectively identifies research trends and detects potential research gaps, assisting users in decision-making.

TABLE II: PERFORMANCE EVALUATION OF RAISA

Metric	Description	Result
Response Time	Time taken to generate the complete report	5–10 sec
Paper Relevance	Accuracy of the retrieved research papers	High
Summary Quality	Clarity and accuracy of AI-generated summaries	High
Gap Detection	Identification of potential research gaps	Effective

D. Comparison Analysis

To test the efficacy of RAISA, a comparison is made with other research-based systems including Google Scholar, Semantic Scholar, and conventional manual techniques.

RAISA exhibits considerable advancements in automation, insight generation, and research gap identification. Whereas conventional methods are mainly concerned with search and extraction, RAISA combines several functions in one software.

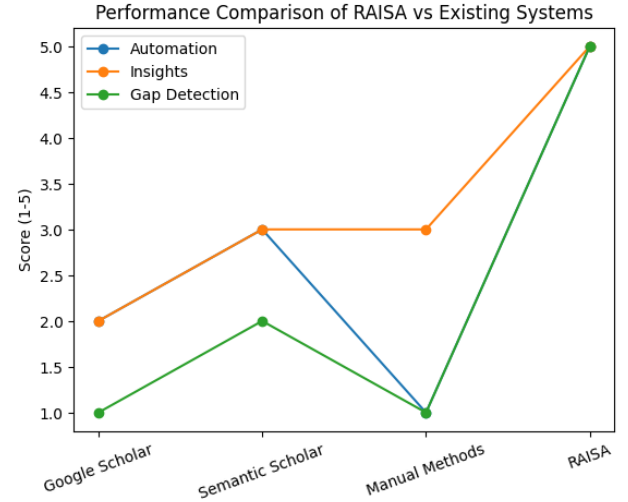


Fig. 6: Performance Comparison of RAISA with Existing Systems

The evaluation metrics, including automation, insight generation, and research gap detection, are measured on a scale of 1 to 5, where higher values indicate better performance.

Fig. 6 shows that RAISA achieves higher performance across all evaluation metrics, including automation, insight generation, and research gap detection.

TABLE III: COMPARATIVE PERFORMANCE OF RAISA VS. OTHER TECHNIQUES

System	Performance Summary
Google Scholar	Low automation, no insight or gap detection
Semantic Scholar	Moderate automation, limited insights and gap detection
Manual Methods	Time-consuming, no automation, limited insights
RAISA	High automation, strong insights, effective gap detection

The comparison clearly indicates that RAISA outperforms traditional systems by integrating multiple functionalities into a single intelligent framework.

E. Report Generation Output



Fig. 7: RAISA pdf first page

VIII. ADVANTAGES

The proposed system offers the following advantages:

- **Automation of research workflow:** Saves researchers time by automating the process of searching, filtering, and analysis.
- **Multi-Agent Design:** Uses agents that perform the tasks of data acquisition, preprocessing, analysis, and reporting separately.
- **Efficient Data Acquisition:** Fetches data about research papers via the OpenAlex API.
- **Advanced NLP Processing:** Uses such processes as keyword extraction via KeyBERT, semantic analysis using Sentence Transformers, and summarization.
- **Research Gap Discovery:** Reveals the gaps that may exist in a field that should be researched more thoroughly.
- **Trends Analysis:** Identifies trends within research over a period of time through analysis of publications.
- **Efficient Reporting:** Generates structured reports in the form of PDF documents with insights and summaries.
- **Intuitive Interface:** Makes it possible to interact with the system through a web interface easily.
- **Increased Efficiency:** Saves a lot of time due to automation of research workflow.
- **Scalability:** Is easy to extend with new agents and functionality.

IX. LIMITATIONS

Despite its effectiveness, the proposed system has certain limitations:

- **Dependence on Third-party APIs:** Data extraction in the current the proposed system is done using an external API provided by OpenAlex. This can result in limited availability and slow responses from APIs, as well as restricted API calls.
- **Incomplete Dataset:** Information obtained through the analyzed dataset can be of low quality if research papers on the specified topic are limited to the available databases.

- **Simple Method of Research Gap Identification:** Current research gaps identification can be improved by analyzing trends and statistics in addition to the most frequent keywords.
- **AI Summary Limitations:** Automatic summarization of original research papers might not reveal certain specific details contained within them.
- **Inability to Process Large Datasets:** Handling many papers at once may be problematic, requiring significant computational power.
- **Non-Learning System:** Currently, there is no continuous learning in the proposed system based on user experience.
- **Generalization:** No personalization is possible yet.

X. FUTURE WORK

Following are some possible future developments that can be done on the proposed system to increase its effectiveness and scalability even further:

- **Advanced Research Gap Detection:** Future work can explore the possibility of using deep learning and knowledge graphs for finding relevant research gaps.
- **Support for Multiple Data Sources:** Extending RAISA to include other academic databases including IEEE Xplore, PubMed, and arXiv for getting diverse papers from different sources.
- **Real-Time Learning and Adaptations:** Implementing machine learning algorithms for RAISA to learn from user feedback and provide more accurate results.
- **Personalized Recommendations:** Using user profiles to give personalized suggestions on topics that may interest a particular user.
- **Optimizations for Scalability:** Improving efficiency of the system by implementing optimizations for processing big data on cloud servers.
- **Advanced Summarization:** Employing more advanced neural architectures to create accurate paper summaries.
- **Workflow Automation Platforms:** In future versions of the platform, n8n integration would make it possible to automate certain tasks.
- **More Visualizations:** Adding more advanced visualizations such as graphs and topic modeling to help understand the output data better.

XI. CONCLUSION

In this paper, we introduced RAISA, a multi-agent research assistant intended for automation of the literature review and research analysis process. In our approach, we used the power of Artificial Intelligence and Natural Language Processing capabilities for performing operations such as data collection, keyword mining, semantic analysis, summarization, trend identification, and identifying the research gap.

The findings suggest that RAISA is successful in automating the time-consuming process of research paper analysis. Moreover, it provides meaningful insights on the analyzed data via dashboard and reporting capabilities of the system. In comparison to existing research frameworks, our approach

presents a more automated and unified way of dealing with numerous analytical tasks.

Finally, we can conclude that our research assistant helps researchers save time and increase their productivity significantly.

The proposed system can serve as a foundation for next-generation intelligent research assistant platforms.

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